Revision Game Cards — 1.2 Software & Software Development

Print and cut along the borders. ✂️ Loop cards: each player reads the bottom "Who has…?"; whoever holds the matching "I have…" reads it out, then their own clue — the chain visits every card and loops back to the start.

Loop cards (dominoes) — cut out & deal

1 / 15

I HAVE

Operating system

WHO HAS…

the OS job of allocating RAM to processes, using techniques like paging and segmentation??

2 / 15

I HAVE

Memory management

WHO HAS…

dividing memory into fixed-size blocks called pages and frames??

3 / 15

I HAVE

Paging

WHO HAS…

dividing memory into variable-size, logical blocks??

4 / 15

I HAVE

Segmentation

WHO HAS…

using secondary storage as if it were RAM to run more or larger programs??

5 / 15

I HAVE

Virtual memory

WHO HAS…

the slowdown caused by excessive swapping of pages between RAM and disk??

6 / 15

I HAVE

Disk thrashing

WHO HAS…

a signal that a device or process needs the processor's attention??

7 / 15

I HAVE

Interrupt

WHO HAS...

the data structure (LIFO) onto which registers are pushed so a job can resume after an interrupt??

8 / 15

I HAVE

Stack

WHO HAS...

the scheduling algorithm giving each process a fixed time slice (quantum) in turn??

9 / 15

I HAVE

Round robin

WHO HAS...

the firmware that runs first at start-up, performs the POST, and loads the OS??

10 / 15

I HAVE

BIOS

WHO HAS...

the translator that turns high-level source into machine code all at once, producing an executable??

11 / 15

I HAVE

Compiler

WHO HAS...

the first stage of compilation, which breaks code into tokens and builds the symbol table??

12 / 15

I HAVE

Lexical analysis

WHO HAS...

the development methodology built around risk analysis at every iteration??

13 / 15

I HAVE

Spiral

WHO HAS...

the OOP feature where a subclass acquires the attributes and methods of a superclass??

14 / 15

I HAVE

Inheritance

WHO HAS...

the addressing mode where the operand is the actual value to be used??

I HAVE

Immediate addressing

WHO HAS…

the software that manages hardware and provides a platform for applications??

Taboo cards — describe the term without the banned words

INTERPRETER

Don't say:

- line
- execute
- run
- slow

COMPILER

Don't say:

- executable
- machine code
- all
- once

VIRTUAL MEMORY

Don't say:

- RAM
- secondary storage
- swap
- disk

INTERRUPT

Don't say:

- signal
- processor
- attention
- ISR

PAGING

Don't say:

- fixed
- blocks
- frames
- memory

ENCAPSULATION

Don't say:

- data
- hiding
- private
- methods

POLYMORPHISM

Don't say:

- method
- different
- object
- override

AGILE

Don't say:

- iterative
- feedback
- change
- increments

Bingo — word bank & ready-to-cut cards

Word bank (students fill a blank 3×3 grid, or use the pre-filled cards below): Operating system, Paging, Segmentation, Virtual memory, Disk thrashing, Interrupt, ISR, Round robin, Real-time OS, BIOS, Device driver, Compiler, Interpreter, Assembler, Linker, Loader, Waterfall, Agile, Encapsulation, Polymorphism

BINGO 1

Assembler	Virtual memory	Interpreter
Compiler	Segmentation	Round robin
Loader	Real-time OS	ISR

BINGO 2

Paging	ISR	Segmentation
Interpreter	Interrupt	Encapsulation
Compiler	Loader	BIOS

BINGO 3

Waterfall	Assembler	Round robin
Disk thrashing	Polymorphism	Loader
Agile	Segmentation	Device driver

BINGO 4

Loader	Agile	Compiler
Round robin	Linker	Waterfall
Real-time OS	Polymorphism	ISR

BINGO 5

Assembler	Virtual memory	Agile
Waterfall	ISR	Interpreter
Polymorphism	Linker	Encapsulation

BINGO 6

Loader	Interpreter	Agile
Assembler	BIOS	Interrupt
Real-time OS	Device driver	Segmentation

Bingo — caller's list (teacher: read the clue, students cross off the term)

#	Clue to read aloud	Answer
1	Software that manages hardware and provides a platform for applications.	Operating system
2	Dividing memory into fixed-size blocks (pages and frames).	Paging
3	Dividing memory into variable-size, logical blocks.	Segmentation
4	Using secondary storage as if it were RAM.	Virtual memory
5	Excessive swapping between RAM and disk that slows the system.	Disk thrashing
6	A signal that a device or process needs the processor's attention.	Interrupt
7	The OS code run to deal with one particular interrupt.	ISR
8	Scheduling that gives each process a fixed time slice in turn.	Round robin
9	An OS that guarantees a response within a set time limit.	Real-time OS
10	Firmware that runs first, performs the POST and loads the OS.	BIOS
11	Software that lets the OS control a specific piece of hardware.	Device driver
12	Translates high-level source into machine code all at once.	Compiler
13	Translates and executes high-level source one line at a time.	Interpreter
14	Translates assembly language into machine code (one-to-one).	Assembler
15	Combines a compiled program with libraries/modules into one executable.	Linker
16	OS software that copies an executable into RAM ready to run.	Loader

Quick-fire — clue & answer (self-test / quiz-quiz-trade)

#	Clue	Answer
1	The data structure (LIFO) used to save registers during an interrupt.	Stack
2	A page is needed but it is currently on disk, so the OS must swap it in.	Page fault
3	Pre-written, pre-compiled, tested subroutines you can reuse in your program.	Library
4	The second stage of compilation, where tokens are checked against the grammar.	Syntax analysis / parsing
5	A scheduling property: a running process can be stopped and the CPU reallocated.	Pre-emptive
6	Platform-independent code (e.g. Java bytecode) run by a virtual machine.	Intermediate code
7	A template/blueprint that defines a class's attributes and methods.	Class
8	A specific instance of a class.	Object
9	The addressing mode where the operand is the address holding the value.	Direct addressing
10	The addressing mode where the operand points to a location holding the address of the value.	Indirect addressing
11	The XP practice of two programmers working at one machine.	Pair programming
12	The methodology built around rapid prototyping refined by user feedback.	RAD